

- Q.26 Write a note on IRC classification of bridges.
 Q.27 Enlist any five necessity of ventilation of tunnels.
 Q.28 Describe the working of "Diamond Crossing" with a neat sketch.
 Q.29 Write the disadvantages of ROB.
 Q.30 Describe the problems arise during the sinking of a well foundation.
 Q.31 Describe the requirements of the ideal tunnels. (any five)
 Q.32 Write the various points to be taken into consideration during the "Inspection of Bridges" required for their maintenance.
 Q.33 Write down the classification of bridges according to their structural form.
 Q.34 Write the various points of differences between a bridge and a culvert.
 Q.35 Describe the purpose of providing the yards in railways stations.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
 Q.36 Describe method of ventilation by combination blowing and exhaust.
 Q.37 Write a short note on :-
 a) Inspection of bridge
 b) Routine Maintenance
 Q.38 Explain the functions of an abutment and write the working of abutments with splayed wing walls.

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5th Sem / Civil, Constr. Mgmt., Civil Engg. (Spl. Highway Engg.) Sub : Railways, Bridges & Tunnels

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The clear gauge distance between two rails of narrow gauge is _____
 a) 1000 mm b) 676mm
 c) 662mm d) 762mm
 Q.2 The wing walls constructed at 45° to the abutments are known as _____
 a) Straight wing wall b) Return wing wall
 c) Splayed wing wall d) Inclined Wing Wall
 Q.3 Drifters can be used to drill _____
 a) Only up holes
 b) Only down holes
 c) Horizontal and up holes
 d) Horizontal down and up holes
 Q.4 The main objective of _____ survey is to carry out the detailed survey along the line.
 a) Preliminary Survey
 b) Traffic Survey
 c) Location Survey
 d) Reconnaissance Survey
 Q.5 The suitable tunneling method used for laying under ground sewer is _____.
 a) Army method b) Needle beam method
 c) Italian method d) German method

- Q.6 Which bearing permits the longitudinal movements of the girders _____
 a) Fixed bearings b) Free bearings
 c) Both (a) & (b) d) None of the above
- Q.7 Bearings are provided in the bridges to allow free movement of the _____ without any difficulty.
 a) Girder b) Piers
 c) Abutment d) End supports
- Q.8 Propulsion method of ventilation is also known as _____
 a) Circulation Method b) Blowing method
 c) Exhausting method d) None of the above
- Q.9 Culverts of length less than one metre are known as _____
 a) Water ways b) Vent ways
 c) Cause ways d) None of the above
- Q.10 A minimum illumination of _____ should be provided at the tunnel and shaft heading during drilling, mucking and sealing.
 a) 50 lux b) 75 lux
 c) 100 lux d) 125 lux

SECTION-B

Note: Objective type questions (Select the appropriate option). All questions are compulsory. (10x1=10)

- Q.11 The walls which are constructed at both ends of the abutments to retain the earth filling of bridge approaches are called _____
- Q.12 The angle between the running faces of point rail and splice rail at a crossing is known as _____ (angle of crossing / Toe of crossing)
- Q.13 Removal of loose rock fragments and hammering off projecting pointed rock is known as _____ (mucking/scaling)

- Q.14 For deciding the _____ of well foundation grip length must be considered as per IRC recommendations. (Width / Depth)
- Q.15 Broken stones type of ballast is most preferred for _____ tracks.
- Q.16 The art of keep the tunnels _____ during the after the construction operations is called drainage of tunnels. (Free from water / full of water)
- Q.17 Circular section of tunnel is _____ for placement of concrete lining. (Suitable / Not suitable)
- Q.18 _____ are the members which are laid under the rails to hold the rails for required gauge as well as to well as to support the rails.
- Q.19 The railways provided at elevated portion, above the ground surface are called _____ (Surface railways / Elevated railways)
- Q.20 Removal of loose rock fragments and hammering off projecting pointed rock is known as _____ (mucking / scaling)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 List five requirements of ideal rail sleepers.
- Q.22 Describe the working of abutments with splayed wing walls.
- Q.23 Write the requirements (any five) for materials to be good ballast.
- Q.24 Describe the working of fixed plate bearings, used in bridges.
- Q.25 Write a short note on "Requirements of Drainage Systems" in railways tracks.